

MIT 5742-100 DATA SCIENCE AND ANALYTICS

Predicting Global Video Game Sales Based on Platform, Genre, and Sales in Different Regions

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Problem Statement

The question we wanted to explore was how global sales were impacted by different factors of video games.

The video game industry is dynamic and ever evolving, with numerous factors influencing global sales trends. Our research delves into understanding the intricate interplay between platform types, game genres, and regional sales patterns to uncover insights that could inform strategic decision-making in the gaming industry. By examining these factors comprehensively, we seek to provide valuable insights that can guide stakeholders in optimizing their product offerings, marketing strategies, and resource allocation to maximize sales and enhance overall market competitiveness.

Description of the Data

Our dataset, sourced from Kaggle, comprises comprehensive data on video game sales, ranked by unit sales in descending order. With 11 columns and 16,600 rows, this dataset offers a rich repository of information for analysis. The dataset encompasses various attributes, including game titles, platforms, genres, publishers, release years, and sales figures. These sales figures are categorized into global sales and regional sales across different geographic markets, measured in millions of units sold.

Data Types:

- **Rank:** Integer
- **Name:** String
- **Platform:** String
- **Year:** Integer
- **Genre:** String
- **Publisher:** String
- **NA_Sales:** Float
- **EU_Sales:** Float
- **JP_Sales:** Float
- **Other_Sales:** Float
- **Global_Sales:** Float

This dataset provides valuable insights for analyzing sales trends, discerning customer preferences, and facilitating strategic decision-making processes for gaming companies within the industry.

Analyses Procedure

Preprocessing: Our dataset, though extensive, posed challenges in analysis. Utilizing Python libraries such as Pandas, SciKit-Learn, and Matplotlib, we initially processed the CSV file by creating dummy variables for Genre and Platform. We opted for this approach due to their manageable unique categorical values compared to Publisher. To address the skewness caused by numerous near-zero Global Sales, we balanced the dataset by sampling an equal number of records above and below the mean. This significantly reduced bias in our model. Pre-stratification, our dataset contained 16,598 rows, with 12,842 classified as "Low" sales and 3,756 as "High." Post-stratification, we had 7,512 rows, evenly distributed between "Low" and "High" categories.

Linear Regression: This report explores the relationship between video game sales in different regions and their impact on global sales figures. A multiple linear regression analysis was conducted using a dataset of 16,598 observations. This analysis aimed to predict global sales based on sales figures from four specific regions: North America (NA), Europe (EU), Japan (JP), and Other (encompassing all other regions).

The resulting regression equation is:

$$\text{Global Sales} = 0.000323 + 0.99 * \text{NA Sales} + 0.99 * \text{EU Sales} + 0.99 * \text{JP Sales} + 0.99 * \text{Other Sales}$$

In Excel, we conducted a linear regression analysis of the data to predict the global sales based on regional sales. This produced a practically perfect model with an r-squared value of 0.99 out of 1. The p-values for most independent variables are all 0. This can be seen in Figure 1. There are very small residuals in the residuals plot, which can be seen in Figure 2. As sales increased, there is not a lot of variability in the residuals. This exemplifies a near perfect model. The line plot for tracking the fit of the model, which can be seen in Figure 3, has points for actual data and predictions. The points of the actual global sales and predicted global sales overlap almost completely. This means that our predictions were almost the exact same as the actual values. Once again, this indicates that the linear model is perfect. However, there are issues with this model that are addressed later in the report under the major results section.

Logistic Regression: The success of a video game or any product can be judged by many factors. However, the simplest outcome for most products to be judged as successful is through sales. Thus, we decided to train a logistic regression model to predict whether a video game will be commercially successful based on the data set's features. The strength of logistic regression falls in its use in binary classification tasks. Our prediction variable of commercial success is a binary True or False Boolean output.

For the inputs, we decided to use the Genre and Platform data to predict Global Sales. After creating the dummy variables addressed in the preprocessing, we split the data into training (2/3) and testing (1/3) sets. Then, we initialized the model using the LogisticRegression() method, fit the model using the fit() method, and made predictions using the predict() method. The results can be seen in Figures 4 and 5.

Decision Trees: In Python, we created a decision tree to make predictions about Global Sales using Platform and Genre as the predictor variables. After the preprocessing, we filtered the data

into X and Y variables which are our independent and dependent variables respectively. Our independent variables consist of the dummy variables for Genre and Platform while our dependent variable is Global Sales. Next, using the Scikit-learn package in Python we split the data into training and test sets specifying the test set size as 0.33 of the data. Next, we used Scikit-learn to import DecisionTreeClassifier to create the decision tree. We created a decision tree object using entropy as the criterion and set the max depth of the tree to 5. We then fit the data to the decision tree object and made predictions. This data was then used to create and visualize a confusion matrix. Using the matrix, we were able to calculate the accuracy, specificity and sensitivity. The results can be seen in Figures 6, 7, 8, 9, and 10.

Association Rules: We utilized Python and libraries like Pandas and MLxtend to extract association rules from our dataset, predicting global sales based on platform, genre, and regional sales data. After loading our dataset into a Pandas DataFrame, we transformed categorical variables like 'Genre' and 'Platform' into binary format using Pandas' `get_dummies()` method for association rule mining. However, we encountered an issue where the multitude of unique platform values overshadowed genre categories. To address this, we filtered the dataset based on platform frequency, focusing on top platforms while preserving genre diversity. We further simplified the data by dropping unnecessary columns and converting sales figures into binary format. Then, employing the Apriori algorithm with a minimum support threshold of 0.1, we mined frequent itemsets and derived association rules, assessing metrics like confidence and lift. These rules unveiled insights into variable relationships, highlighting patterns such as platform-genre associations and regional sales correlations.

Major Results

Linear Regression: Here, the intercept (0.000323) is close to zero, indicating that even with zero sales in all regions, a small global sale is predicted. However, this value is likely statistically insignificant and can be disregarded for practical purposes.

Since global sales are already the sum of sales from the 4 regions, the coefficients of 0.99 for each regional sales variable does not indicate an independent contribution of almost one unit to the global sales figure. Instead, they represent the proportion of their own contribution within the sum. Imagine each regional sales figure contributes a "slice" to the global sales "pie." The coefficients of 0.99 suggest that each "slice" is nearly the same size, meaning each region contributes proportionally almost equally to the overall pie. However, the absolute size of their contribution depends on the actual sales figures in each region.

The multiple R of 0.99 indicates an exceptionally strong positive linear relationship between regional sales and global sales. Similarly, the R-squared value of 0.99 suggests that 99% of the variation in global sales can be explained by the combined effect of sales figures from these four regions.

Furthermore, while the p-value of 0 for the overall regression signifies a statistically significant model, interpreting individual p-values for each variable might be challenging due to potential multicollinearity, which refers to a high correlation between the independent variables (regional sales). This interdependency can make it difficult to isolate the unique effect of each region on global sales and should be considered when drawing conclusions about individual coefficients.

Logistic Regression: The logistic regression model had an accuracy score of 0.625, which is close to the generally accepted minimum of 0.7. Our model does a decent job of predicting the success of video games based on the Genre and Platform for the data set. The specificity was 0.581, which is slightly lower than the accuracy score. The sensitivity was 0.668, which is

slightly higher than the accuracy score. This means that our model might be slightly better at predicting positive outcomes than negative outcomes. The confusion matrix for the data and the metrics can be seen in Figures 4 and 5.

Decision Tree: The decision tree model had an accuracy score of 0.581, which can be considered low considering it is not that far from being random chance (0.5). This model does not do a great job at making correct video game success predictions based on the Genre and Platform, however, it is better than random chance. The specificity was 0.864, which is a lot higher than the accuracy score. The sensitivity was 0.298 which is a lot lower than the accuracy score. These differences indicate that this model is better at predicting negative outcomes (Low sales) than positive outcomes (High sales). The confusion matrix for the data and the metrics can be seen in Figures 7, 8, 9, and 10.

Association Rules: One notable association rule in the analysis is (NA_Sales, Platform_PS2) -> (GenreID, Other_Sales, EU_Sales, Global_Sales). It signifies a strong relationship between North American sales (NA_Sales) for games on the PS2 platform (Platform_PS2) and other variables like GenreID, Other_Sales, EU_Sales, and Global_Sales, with a high lift value of 1.758867 indicating a positive association. A confidence value of 0.930825 suggests a high likelihood of these variables co-occurring, supported by a frequent occurrence (support value: 0.107100). The positive leverage value (0.046209) indicates NA_Sales and Platform_PS2 contribute to the overall presence of consequent variables. These insights are crucial for strategic decision-making in the gaming industry, enabling businesses to tailor their strategies and offerings to customers preferring NA_Sales and Platform_PS2.

Conclusions

In our analysis, we noted that global sales were heavily skewed towards zero, impacting our metrics and analysis outcomes. To address this, we stratified the dataset, used a smaller sample, and split sales based on the mean. However, our models still struggled to predict commercial success accurately. Despite this, our goal was to understand the video game industry by analyzing sales influencers. We utilized various methods like linear regression, logistic regression, decision trees, and association rules. The logistic regression achieved moderate success with an accuracy of 0.625, while decision trees performed slightly better than chance at 0.581. Association rules revealed a strong link between PS2 platform sales in North America and specific genres, highlighting the importance of platform-specific strategies. Though our linear regression showed a near-perfect fit, it's influenced by dataset skewness. While our analysis offers insights, more data and improved methods could enhance predictions and benefit industry stakeholders.

References

The Devestator. (2023). Global Video Game Sales [Data set]. Kaggle.

<https://www.kaggle.com/datasets/thedevastator/global-video-game-sales?resource=download>

APPENDICES

Figure 1.

SUMMARY OUTPUT									
Regression Statistics		Global Sales=3.2*10 ⁻⁴ +9.99*10 ⁻¹ *NA Sales+9.99*10 ⁻¹ *EU Sales+9.99*10 ⁻¹ *JP Sales+9.99*10 ⁻¹ *Other Sales							
Multiple R	0.999994362								
R Square	0.999988725								
Adjusted R Square	0.999988722								
Standard Error	0.005222199								
Observations	16598								
ANOVA									
		df	SS	MS	F	Significance F			
Regression		4	40132.95037	10033.23759	367903735.8	0			
Residual		16593	0.452513783	2.72714E-05					
Total		16597	40133.40288						
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%	
Intercept	0.00032295	4.29613E-05	7.517228137	5.88001E-14	0.000238741	0.00041	0.00024	0.00041	
NA_Sales	0.999940582	8.04843E-05	12424.0419	0	0.999782824	1.0001	0.99978	1.0001	
EU_Sales	0.999987583	0.000145226	6885.753928	0	0.999702925	1.00027	0.9997	1.00027	
JP_Sales	0.999883816	0.000149083	6706.909565	0	0.999591598	1.00018	0.99959	1.00018	
Other_Sales	0.99958749	0.00031867	3136.748585	0	0.998962863	1.00021	0.99896	1.00021	

Figure 2.



Figure 3.

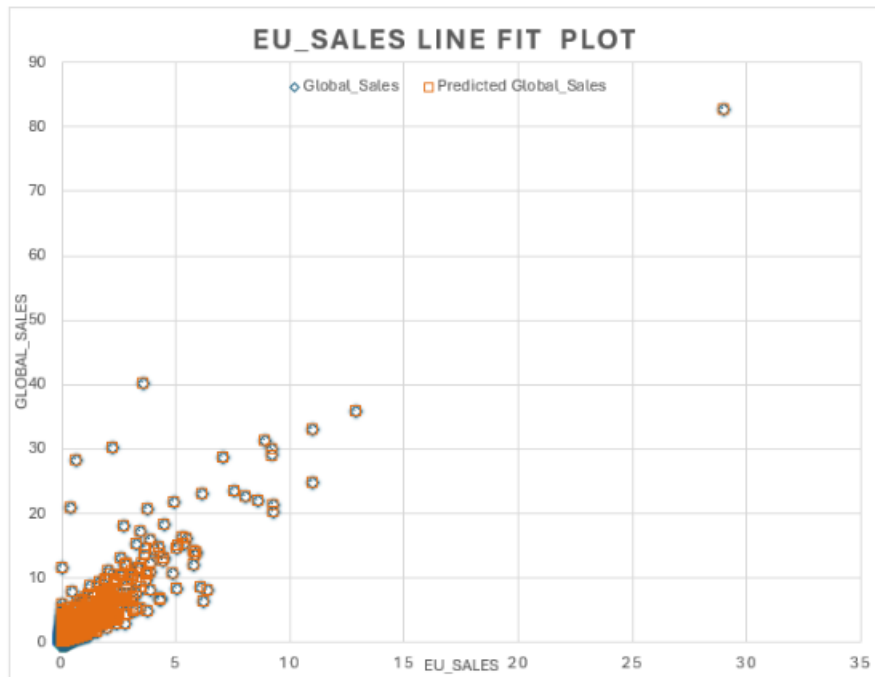


Figure 4.

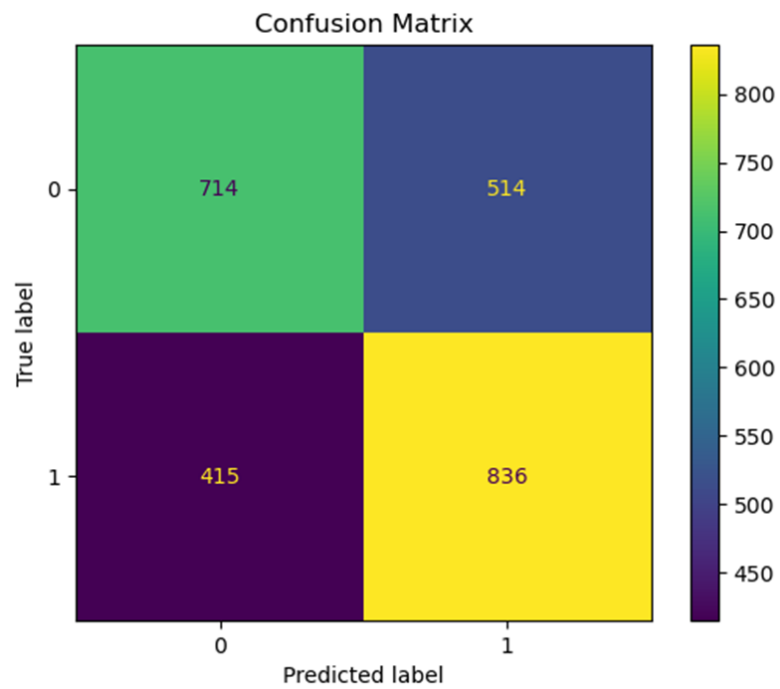


Figure 5.

```

In [30]: > TN, FP, FN, TP = conf_matrix.ravel()

In [31]: > accuracy = accuracy_score(y_test, predictions)
print("Accuracy:", accuracy)

Accuracy: 0.6252521177894312

In [32]: > # Calculate specificity

specificity = TN / (TN + FP)
specificity

Out[32]: 0.5814332247557004

In [33]: > # Calculate sensitivity

sensitivity = TP / (TP + FN)
sensitivity

Out[33]: 0.6682653876898481

```

Figure 6.

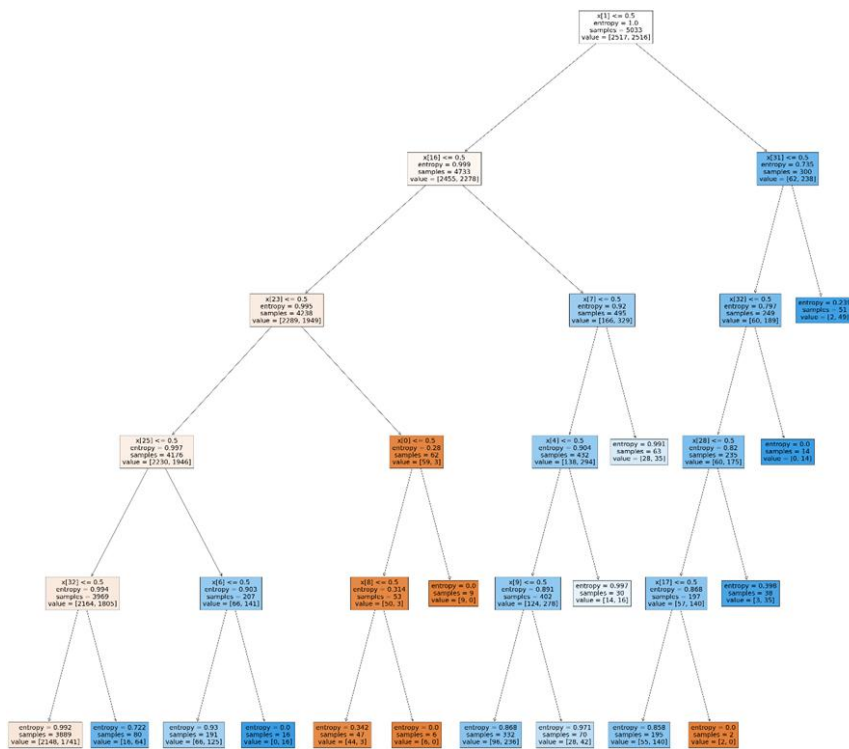


Figure 7.

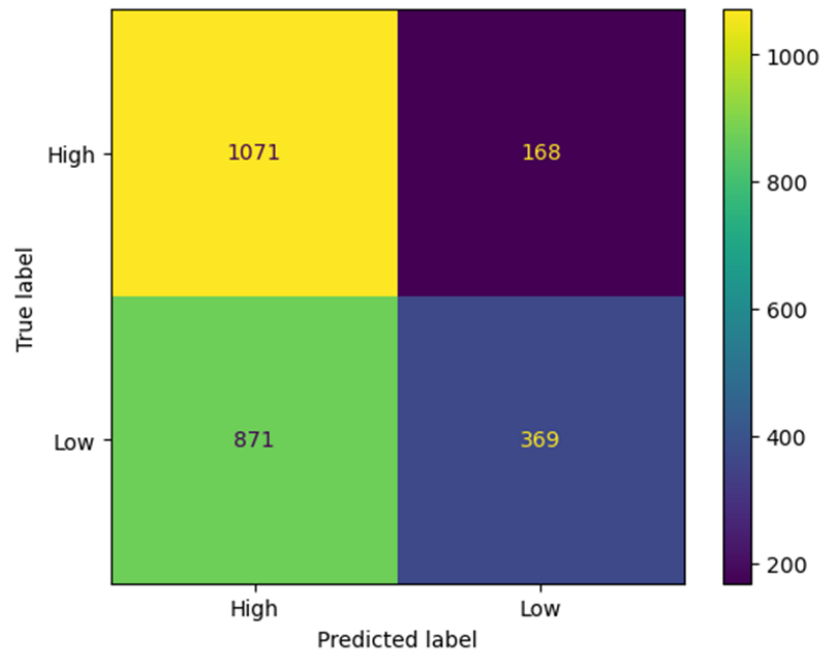


Figure 8.

```
In [21]: 1 # Step 4: We assess the accuracy of our predictions
          2
          3 from sklearn.metrics import accuracy_score
          4 performance_of_model=accuracy_score(Y_test, Y_pred)
          5 performance_of_model
```

Out[21]: 0.5808793868495361

Figure 9.

```
In [71]: 1 # Calculate specificity
          2
          3 specificity = TN / (TN + FP)
          4 specificity
```

Out[71]: 0.864406779661017

Figure 10.

```
In [72]: 1 # Calculate sensitivity
          2
          3 sensitivity = TP / (TP + FN)
          4 sensitivity
```

Out[72]: 0.29758064516129035

Figure 11.

```
In [78]: resultDF=rules[(rules['lift']>=1.5)&(rules['confidence']>=0.6)]
```

```
In [79]: resultDF
```

Out[79]:

	antecedents	consequents	antecedent support	consequent support	support	confidence	lift	leverage	conviction	zhangs_metric
94	(JP_Sales, NA_Sales)	(Other_Sales)	0.150178	0.633945	0.143825	0.957694	1.510689	0.048620	8.652565	0.397789
173	(JP_Sales, EU_Sales)	(Other_Sales)	0.140054	0.633945	0.135237	0.965603	1.523165	0.046450	10.642109	0.399411
198	(Platform_PS2, EU_Sales)	(Other_Sales)	0.114152	0.633945	0.108567	0.951070	1.500240	0.036200	7.481241	0.376408
316	(JP_Sales, NA_Sales, EU_Sales)	(Other_Sales)	0.135865	0.633945	0.132863	0.977903	1.542567	0.046732	16.566099	0.407032
319	(JP_Sales, NA_Sales)	(Other_Sales, EU_Sales)	0.150178	0.529219	0.132863	0.884705	1.671719	0.053386	4.083268	0.472820
...	...	...	...	...	...	...	...	...	...	...
1391	(Other_Sales, Global_Sales, Platform_PS2)	(GenreID, NA_Sales, EU_Sales)	0.110731	0.596593	0.107100	0.967213	1.621228	0.041039	12.303917	0.430897
1392	(GenreID, Other_Sales, Platform_PS2)	(NA_Sales, EU_Sales, Global_Sales)	0.110731	0.596593	0.107100	0.967213	1.621228	0.041039	12.303917	0.430897
1397	(NA_Sales, Platform_PS2)	(GenreID, Other_Sales, EU_Sales, Global_Sales)	0.115060	0.529219	0.107100	0.930825	1.758867	0.046209	6.805680	0.487549
1401	(Platform_PS2, EU_Sales)	(GenreID, NA_Sales, Global_Sales, Other_Sales)	0.114152	0.584584	0.107100	0.938226	1.604946	0.040369	6.724800	0.425498
1404	(Other_Sales, Platform_PS2)	(GenreID, NA_Sales, EU_Sales, Global_Sales)	0.110731	0.596593	0.107100	0.967213	1.621228	0.041039	12.303917	0.430897

81 rows x 10 columns